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Module 7 Write Up

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Introduction

This week, we are delving deeper into web security. In this write-up, I will cover intriguing topics that I learned about during the lecture.

Computer Crime laws

Knowingly accessing or using a computer or network (or attempting to do so) for the purpose of fraud; to obtain money, property or services; or to commit theft of proprietary information.

Knowingly and without authorization altering, destroying or damaging any computer, network, software data ect. (or attempting to do so).

Knowingly and without authorization using or accessing a computer or network (or attempting to do so).

The first two offences can be perused as class c felony. The last item is a Class A misdemeanor.

Web Delivery

The internet is mostly based on HTTP traffic but also contains many like the following

- Content - email, html.
- Search Engine - Google, Archie.
- Browsers - DOS Houdini, Mosaic.
- World Wide web - 1990 Http
- Internet - 1975 TCP/IP
- Networks Ethernet
- Computers - Z3, Apple.

Web Browser

The web browser has many different layers because the web browser performs a lot of different functions. Below are a few layers of the browser.

Network Protocol, this is where most of communication is done.

DOM - Document Object Model. This is the part that the user sees.

In this system there are many different injection points into the browser. Below are some of the ways in which a threat may get injected into the browser.

- De-Obfuscated Content - Browser/Extensions
- JavaScript - Script Engine
- HTML - Dom Tree Browser/Extension
- HTML [RAW html] - Winlog ETW/ETL
- HTTP - Network Layer

User Vulnerabilities

Users often engage in risky actions that could compromise security, such as clicking on dubious links or intentionally downloading unverified software. As a result, users represent the weakest link in the security chain.

There are several common strategies used to target users:

- **Phishing**
- **SEO Poisoning**
- **Fake Antivirus Software**
- **Social Media Link Insertion**
- **Forum Link Insertion**

- **Malvertising**

Today, hackers often prioritize tricking users into providing sensitive information, such as bank details or Social Security numbers, rather than installing malware on their devices. This approach allows attackers to bypass the need for malware and still achieve their goals. There are several different ways in which users may be attacked. Below is a list of different strategies for attacking the users.

SEO Poisoning

In this attack, hackers exploit search engine optimization (SEO) techniques, such as Google Trends, to broaden their reach. They manipulate search results to draw users to their sites by leveraging popular topics, celebrities, current events, or educational content. By gaming the search engine's relevance algorithms, they ensure their malicious links appear in place of genuine search results. Although the initial site may appear harmless, clicking on a link can redirect users to malicious content.

Fake Antivirus (AV) Software

Fake antivirus software is a common attack vector that exploits social engineering. This can occur either directly on a computer or via the web. These fraudulent programs often imitate the appearance of legitimate antivirus software, mimicking either Windows' own security features or well-known AV brands. This tactic preys on users' trust in established security tools, making it an effective method for deceiving individuals into installing malicious software.

Lookalike URLs

Attackers use URLs that closely resemble legitimate ones to deceive users. For example, a URL might look like "microsoft" but use an "rn" instead of "m", or "<http://oregonstate.edu>" might be slightly altered to "http://oregonstate.eclu". They sometimes use visually similar characters to further disguise their malicious intent. Although some of these tactics have been mitigated, they have been effective in the past.

Social Media Attacks

Social media platforms have become a prime target for malware distribution and information gathering. Malicious actors use social media to deliver harmful URLs and gather intelligence. Techniques like catfishing, where attackers impersonate friends to collect personal information, are common and exploit users' trust in social connections.

Malvertising

Malvertising involves using advertising networks to deliver malware. Attackers place malicious ads through these networks, targeting specific users or audiences. This method is efficient because it leverages the advertising network to reach victims without needing direct interaction. The attackers may invest in placing their ads strategically to maximize impact.

Watering Hole Attacks

In a watering hole attack, attackers gather intelligence to identify websites frequently visited by their targets. They then inject malicious code into these sites. When targeted users visit the compromised sites, the malware is delivered to their machines, which can subsequently infect their networks. This method is often used against developers and other specific groups.

Mitigation Strategies

- URL/Domain Reputation Systems
- Site Certification Services
- Client and Gateway Antivirus/Antimalware
- Safe URL Shorteners
- Content Provider Education
- End User Education

Browser Security Features

Modern browsers have significantly improved their security features compared to those from even just a few years ago. These enhancements include:

- Same Origin Policy: Prevents scripts from different origins from interacting with each other.
- URL Scheme Access Rules: Regulates how URLs are accessed.
- OS Isolation and Sandboxing: Limits the browser's ability to access the underlying operating system.
- Redirection Restrictions: Controls and prevents malicious redirections.
- URL Reputation Clients and Content Sniffing: Assesses URL safety and content validity.

Browser Security Exploits

Despite these advancements, browsers can still be exploited. Common methods include:

- Malicious Web Content: Downloading and executing harmful content.
- Zero-Day Exploits: Attacks exploiting previously unknown vulnerabilities.
- Code Obfuscation: Attackers use techniques like minification to hide malicious code, making it harder to detect.

DNS Spoofing

DNS spoofing occurs when a DNS server's cache is compromised. The attacker has pre-configured a malicious redirect in the cache. When a user requests legitimate content, they are instead directed to malicious sites. This type of attack is particularly insidious because it misleads users into believing they are accessing safe content.

Appendix

This write-up follows the MLA standard for expository essays.